

Packet Tracer - WLAN Configuration

นายจิรภัทร แก้วดี 673380577-9 Sec2

Addressing Table

Device	Interface	IP Address
Home Wireless Router	Internet	DHCP
<i>Home Wireless Router</i>	LAN	192.168.6.1/27
RTR-1	G0/0/0.2	192.168.2.1/24
<i>RTR-1</i>	G0/0/0.5	192.168.5.1/24
<i>RTR-1</i>	G0/0/0.100	192.168.100.1/24
<i>RTR-1</i>	G0/0/1	10.6.0.1/24
SW1	VLAN 200	192.168.100.100/24
LAP-1	G0	DHCP
WLC-1	Management	192.168.100.254/24
RADIUS Server	NIC	10.6.0.254/24
Home Admin	NIC	DHCP
Enterprise Admin	NIC	192.168.100.200/24
Web Server	NIC	203.0.113.78/24
DNS Server	NIC	10.100.100.252
Laptop	NIC	DHCP
Tablet PC	Wireless0	DHCP
Smartphone	Wireless0	DHCP
Wireless Host 1	Wireless0	DHCP
Wireless Host 2	Wireless0	DHCP

Blank Line - no additional information

WLAN Information

WLAN	SSID	Authentication	Username	Password
Home Network	HomeSSID	WPA2-Personal	N/A	Cisco123
WLAN VLAN 2	SSID-2	WPA-2 Personal	N/A	Cisco123
WLAN VLAN 5	SSID-5	WPA-2 Enterprise	userWLAN5	userW5pass

Blank Line - no additional information

Note: It is not a good practice to reuse passwords as is done in this activity. Passwords have been reused to make it easier to work through the tasks.

Objectives

In this activity, you will configure both a wireless home router and a WLC-based network. You will implement both WPA2-PSK and WPA2-Enterprise security.

- = Configure a home router to provide Wi-Fi connectivity to a variety of devices.
- = Configure WPA2-PSK security on a home router.
- = Configure interfaces on a WLC.
- = Configure WLANs on a WLC.
- = Configure WPA2-PSK security on a WLAN and connect hosts to WLAN.
- = Configure WPA2-Enterprise on a WLAN and connect hosts to the WLAN.
- = Verify connectivity WLAN connectivity.

Background / Scenario

You will apply your WLAN skills and knowledge by configuring a home wireless router and an enterprise WLC. You will implement both WPA2-PSK and WPA2-Enterprise security. Finally, you will connect hosts to each WLAN and verify connectivity.

Instructions

Part 1: Configure a Home Wireless Router.

You are installing a new home wireless router at a friend's house. You will need to change settings on the router to enhance security and meet your friend's requirements.

Step 1: Change DHCP settings.

- a. Open the Home Wireless Router GUI and change the router IP and DHCP settings according to the information in the Addressing Table.
- b. Permit a maximum of **20** addresses to be issued by the router.
- c. Configure the DHCP server to start with IP address **.3** of the LAN network.

d. Configure the internet interface of the router to receive its IP address over DHCP.

Question:

Verify the address. What address did it receive?

e. Configure the static DNS server to the address in the Addressing Table.

The screenshot shows the 'Home Wireless Router' configuration interface, specifically the 'Network Setup' tab. The interface is divided into several sections:

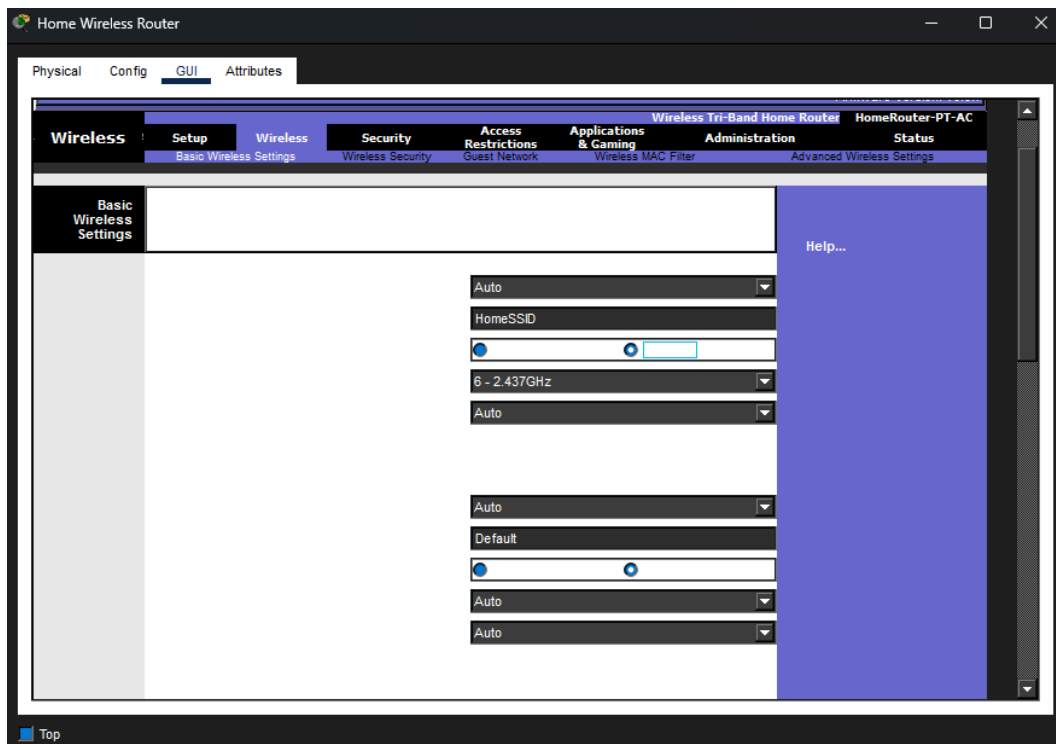
- Optional Settings (required by some internet service providers):** This section contains fields for DNS servers (all set to 0) and a field for the MTU (set to 1500).
- Network Setup:** This section contains the main network configuration options.
 - Router IP:** A dropdown menu showing the IP address 192.168.6.1 and the subnet mask 255.255.255.224.
 - DHCP Server Settings:** This section includes a checkbox for 'DHCP Reservation' (checked), a field for the number of reservations (set to 3), and a field for the reservation period (set to 20).
 - Vlan IDs:** A section for configuring VLANs, with fields for Internet, VoIP, and IPTV.
 - Port Vlan:** A section for configuring VLANs for the ports, with fields for Port 1, Port 2, Port 3, and Port 4.

Step 2: Configure the Wireless LAN.

a. The network will use the 2.4GHz Wireless LAN interface. Configure the interface with the SSID shown in the Wireless LAN information table.

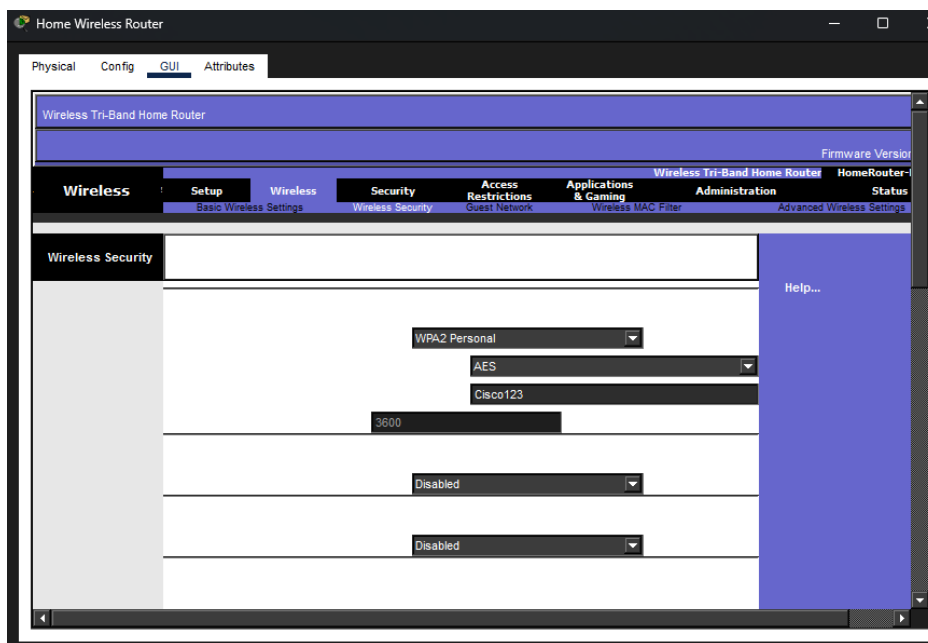
b. Use **channel 6**.

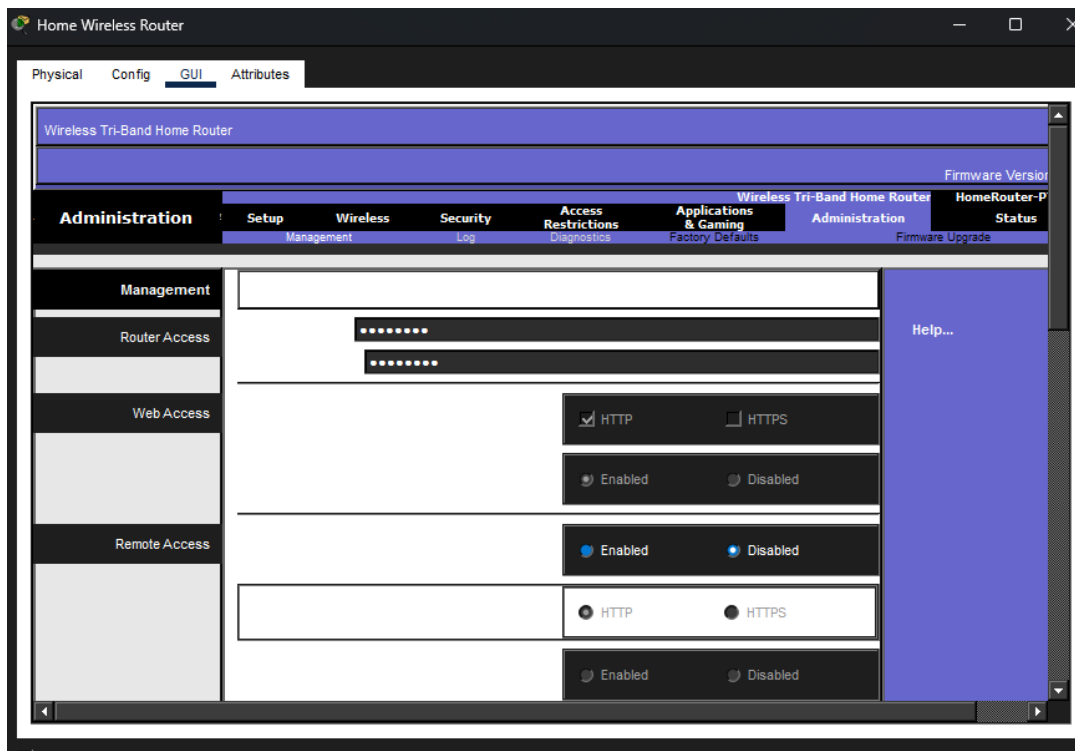
c. Be sure that all wireless hosts in the home will be able to see the SSID.



Step 3: Configure security.

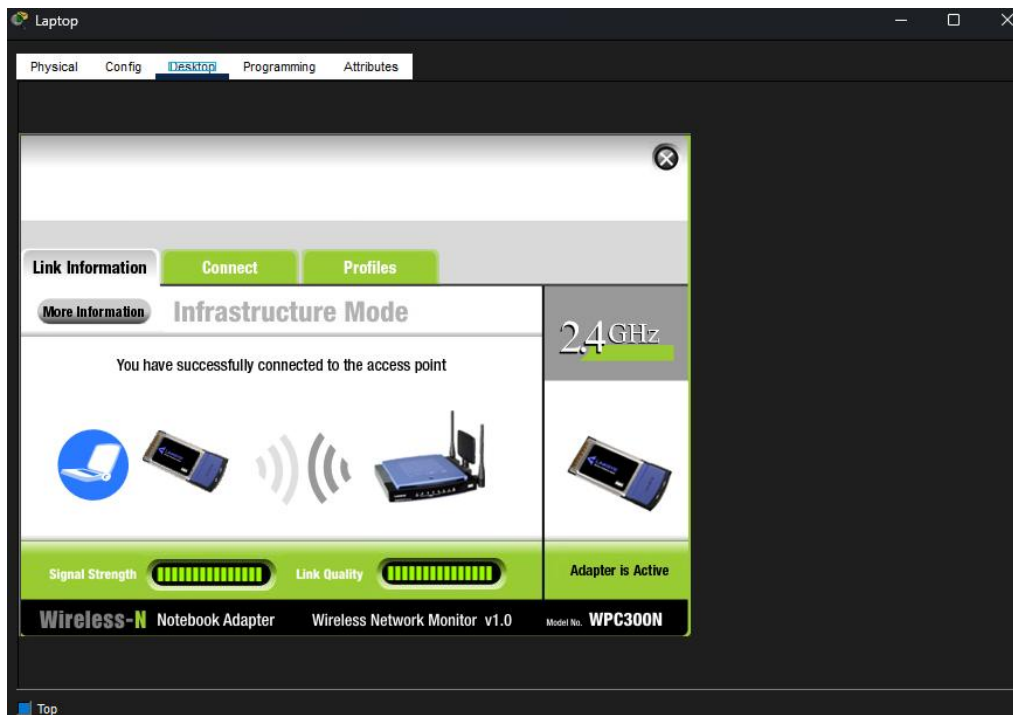
- Configure wireless LAN security. Use WPA2 Personal and the passphrase shown in the Wireless LAN information table.
- Secure the router by changing the default password to the value shown in the Wireless LAN information table.





Step 4: Connect clients to the network.

- Open the PC Wireless app on the desktop of the laptop and configure the client to connect to the network.
- Open the Config tab on the Tablet PC and Smartphone and configure the wireless interfaces to connect to the wireless network.
- Verify connectivity. The hosts should be able to ping each other and the web server. They should also be able to reach the web server URL.



มีการเชื่อมต่อกัน แสดงว่าตั้งค่าถูกต้อง

```
C:\>ping 203.0.113.78

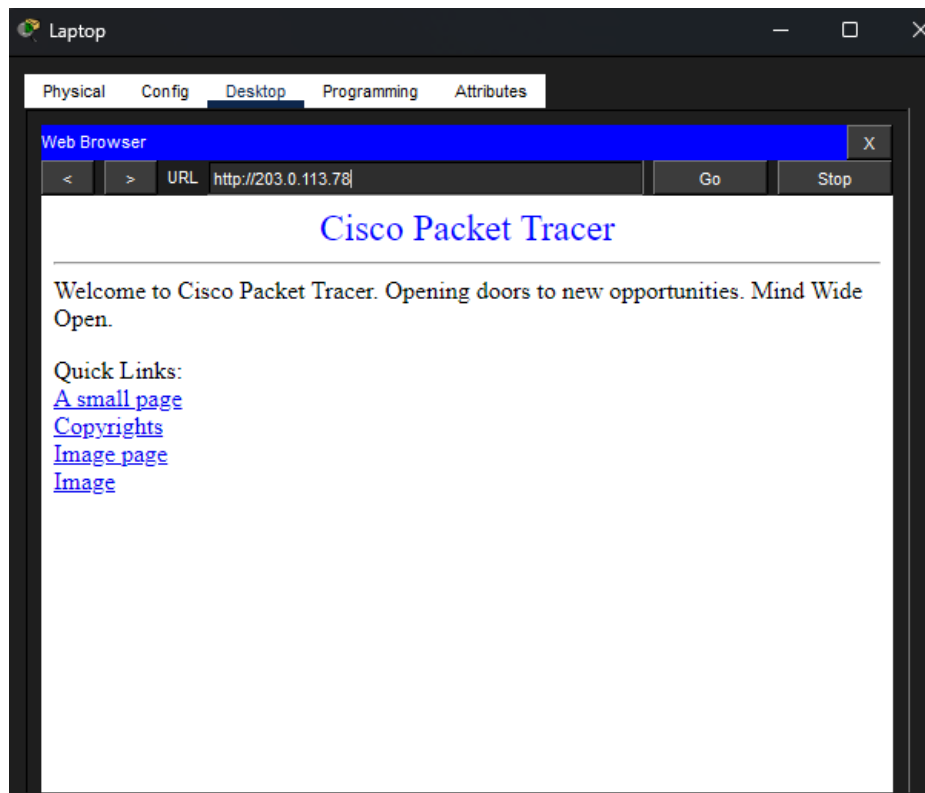
Pinging 203.0.113.78 with 32 bytes of data:

Reply from 203.0.113.78: bytes=32 time=23ms TTL=126
Reply from 203.0.113.78: bytes=32 time=21ms TTL=126
Reply from 203.0.113.78: bytes=32 time=24ms TTL=126
Reply from 203.0.113.78: bytes=32 time=27ms TTL=126

Ping statistics for 203.0.113.78:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 21ms, Maximum = 27ms, Average = 23ms

C:\>
```

มีการตอบกลับแสดงว่าตั้งค่าถูกต้อง



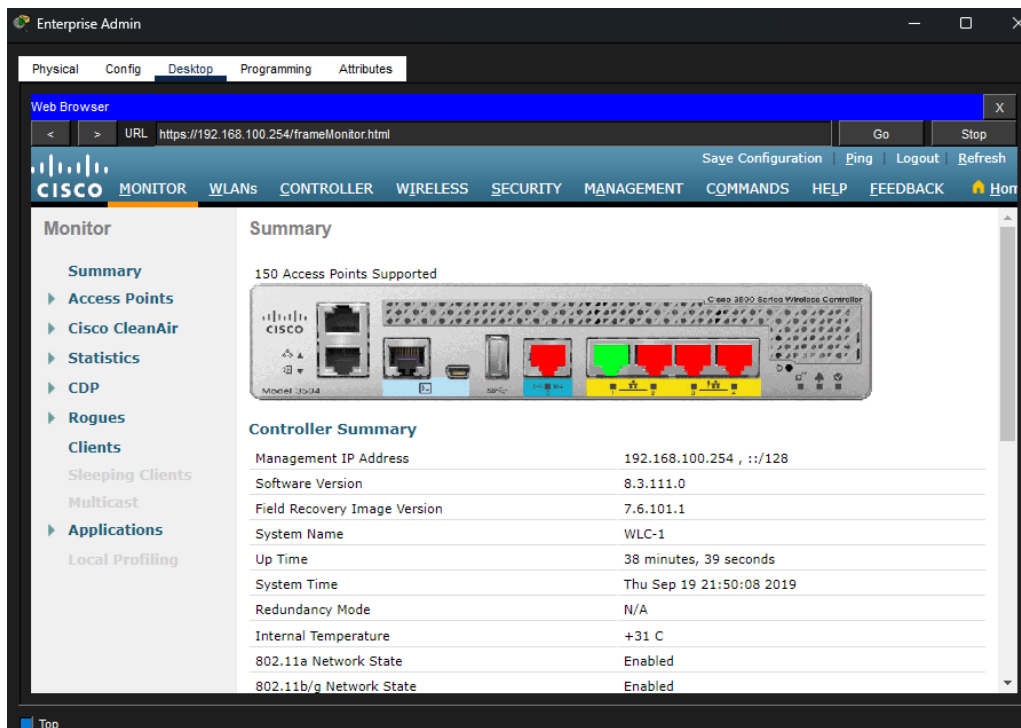
และสามารถเข้าเว็บได้

Part 2: Configure a WLC Controller Network

Configure the wireless LAN controller with two WLANs. One WLAN will use WPA2-PSK authentication. The other WLAN will use WPA2-Enterprise authentication. You will also configure the WLC to use an SNMP server and configure a DHCP scope that will be used by the wireless management network.

Step 1: Configure VLAN interfaces.

- a. From the Enterprise Admin, navigate to the WLC-1 management interface via a web browser. To log into WLC-1, use **admin** as the username and **Cisco123** as the password.



เข้าไปที่หน้าเว็บเพื่อ Configure a WLC Controller Network

b. Configure an interface for the first WLAN.

Name: **WLAN 2**

VLAN Identifier: **2**

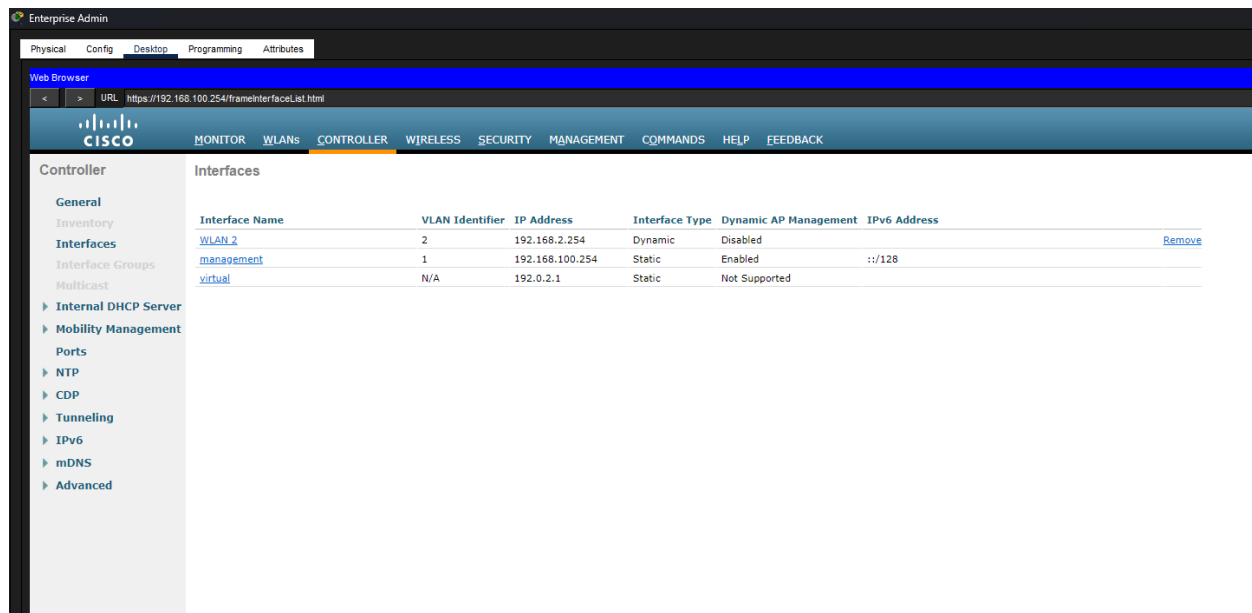
Port Number: **1**

Interface IP Address: **192.168.2.254**

Netmask: **255.255.255.0**

Gateway: **RTR-1 G0/0/0.2 address**

Primary DHCP Server: **Gateway address**



นำตั้งค่ารายละเอียดของ WLAN 2 ให้กรอกข้อมูลตามนี้:

Port Number: 1

IP Address: 192.168.2.254

Netmask: 255.255.255.0

Gateway: 192.168.2.1 (นำมาจาก IP ของ RTR-1 G0/0/0.2)

Primary DHCP Server: 192.168.2.1

c. Configure an interface for the second WLAN.

Name: WLAN 5

VLAN Identifier: 5

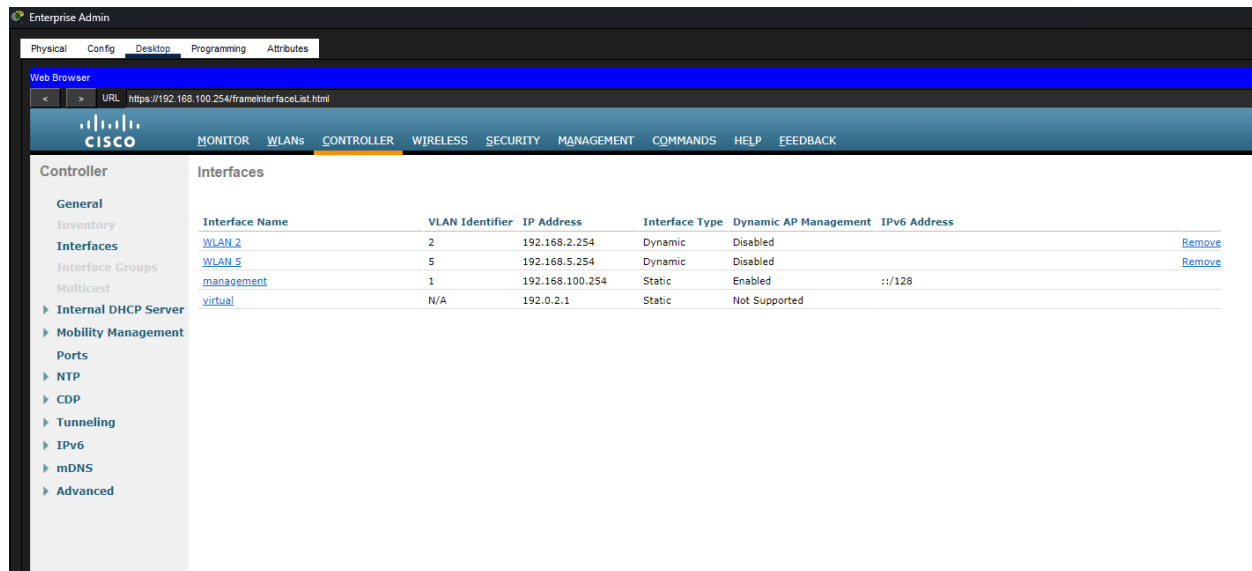
Port Number: 1

Interface IP Address: 192.168.5.254

Netmask: 255.255.255.0

Gateway: RTR-1 interface G0/0/0.5 address

Primary DHCP Server: Gateway address



ละเอียดของ WLAN 5 ตามนี้:

Port Number: 1

IP Address: 192.168.5.254

Netmask: 255.255.255.0

Gateway: 192.168.5.1 (นำมาจาก IP ของ RTR-1 G0/0/0.5)

Primary DHCP Server: 192.168.5.1

Step 2: Configure a DHCP scope for the wireless management network.

Configure and enable an internal DHCP scope as follows:

Scope Name: **management**

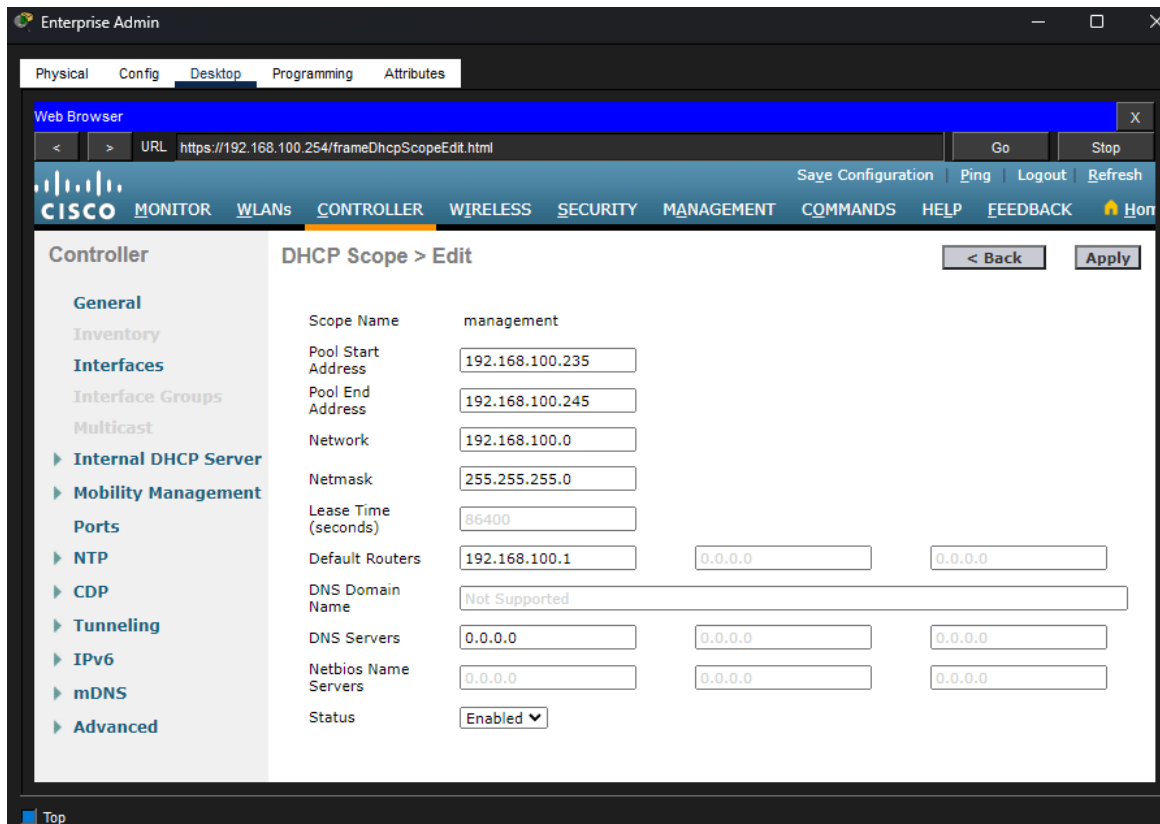
Pool Start Address: **192.168.100.235**

Pool End Address: **192.168.100.245**

Network: **192.168.100.0**

Netmask: **255.255.255.0**

Default Routers: **192.168.100.1**



นำข้อมูลจากโจทย์มากรอกลงในช่องต่างๆ ให้ครบถ้วน ดังนี้:

- Pool Start Address: 192.168.100.235
- Pool End Address: 192.168.100.245
- Network: 192.168.100.0
- Netmask: 255.255.255.0
- Default Routers: 192.168.100.1

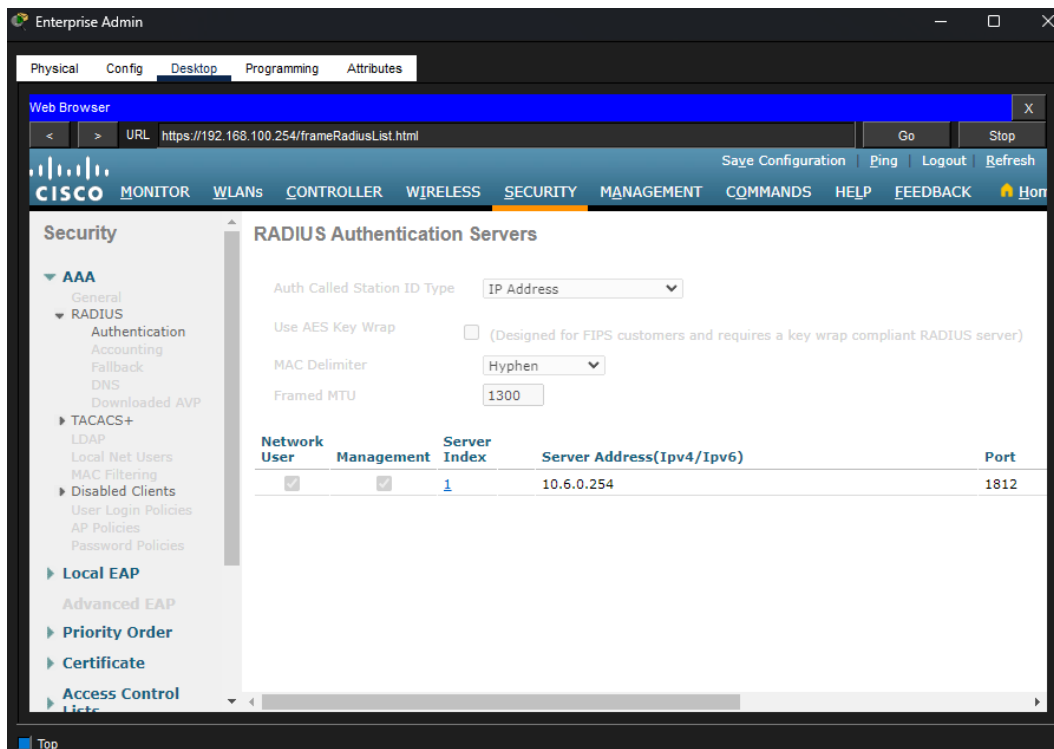
Step 3: Configure the WLC with external server addresses.

a. Configure the RADIUS server information as follows:

Sever Index: **1**

Sever Address: **10.6.0.254**

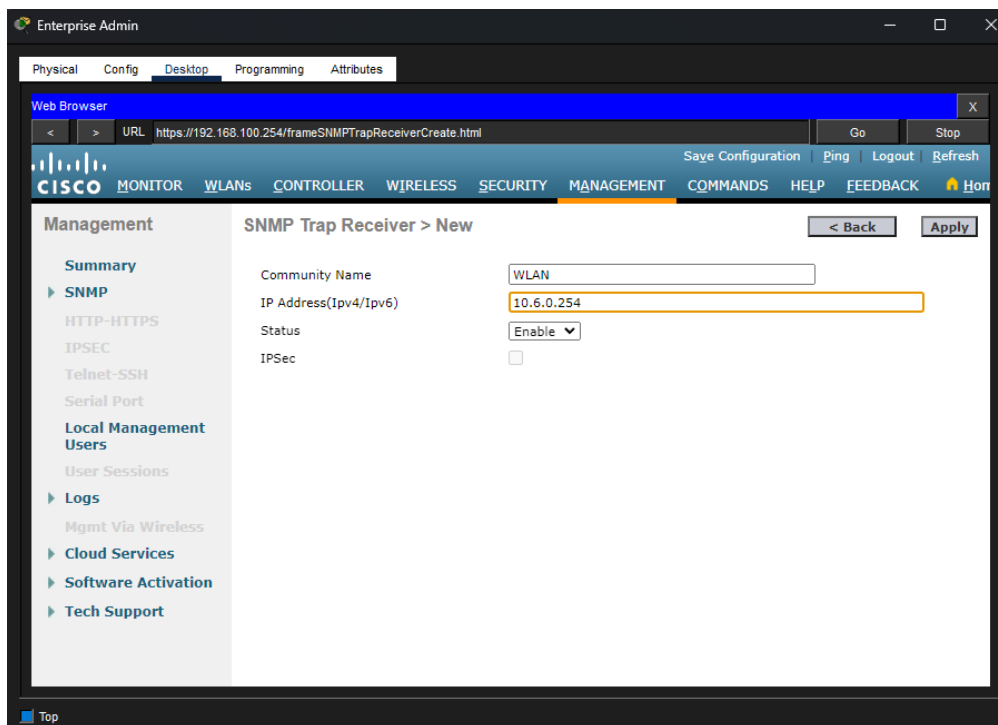
Shared Secret: **RadiusPW**



b. Configure the WLC to send logs information to an SNMP server.

Community Name: **WLAN**

IP Address: **10.6.0.254**



Step 4: Create the WLANs.

a. Create the first WLAN:

Profile Name: **Wireless VLAN 2**

WLAN SSID: **SSID-2**

ID: **2**

Interface: **WLAN 2**

Security: **WPA2-PSK**

Passphrase: **Cisco123**

Under the Advanced tab, go to the FlexConnect section. Enable **FlexConnect Local Switching** and **FlexConnect Local Auth**.

☐	WLAN ID	Type	Profile Name	WLAN SSID	Admin Status	Security Policies	
☐	2	WLAN	Wireless VLAN 2	2	Enabled	[WPA2][Auth(PSK)]	Remove

ทำการตั้งค่าตามโจทย์

b. Create the second WLAN:

Profile Name: **Wireless VLAN 5**

WLAN SSID: **SSID-5**

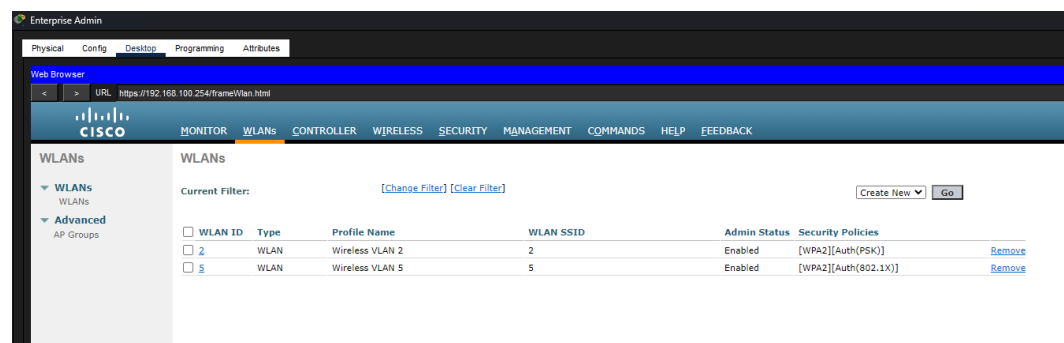
Interface: **WLAN 5**

ID: **5**

Security: **802.1x - WPA2-Enterprise**

Configure the WLAN to use the RADIUS server for authentication.

Make the **FlexConnect** settings as was done in Step 4a.



ทำการตั้งค่าตามโจทย์

Step 5: Configure the hosts to connect to the WLANs.

Use the desktop PC Wireless app to configure the hosts as follows:

- a. Wireless Host 1 should connect to Wireless VLAN 2.



เชื่อมต่อได้

- b. Wireless Host 2 should connect to Wireless VLAN 5 using the credentials in the WLAN information table.



เชื่อมต่อได้

Step 6: Test connectivity.

Test connectivity between the wireless hosts and the Web Server by ping and URL.

```
Wireless Host 1
Cisco Packet Tracer PC Command Line 1.0
C:\>ping 203.0.113.78

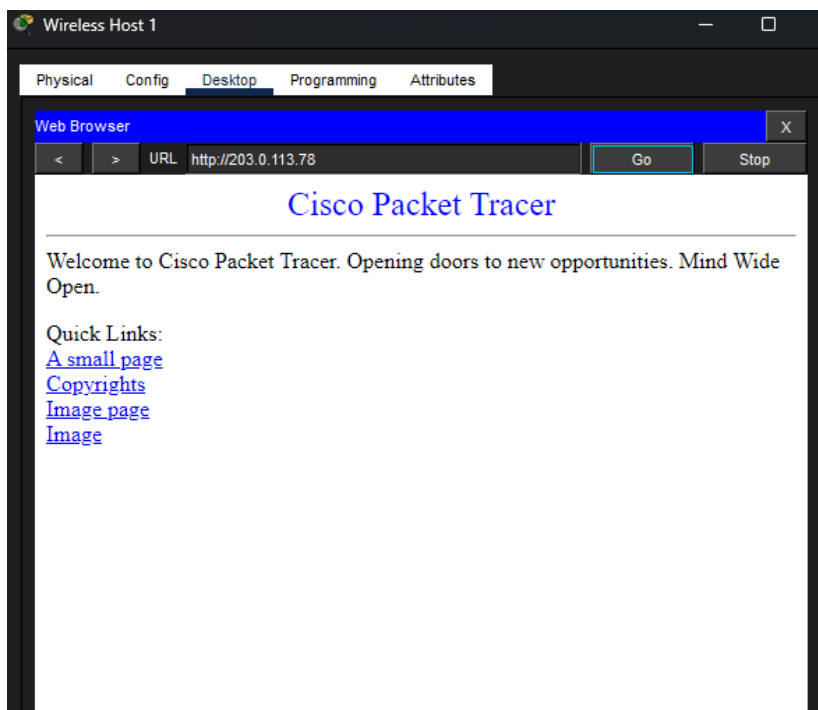
Pinging 203.0.113.78 with 32 bytes of data:

Reply from 203.0.113.78: bytes=32 time=36ms TTL=126
Reply from 203.0.113.78: bytes=32 time=13ms TTL=126
Reply from 203.0.113.78: bytes=32 time=16ms TTL=126
Reply from 203.0.113.78: bytes=32 time=24ms TTL=126

Ping statistics for 203.0.113.78:
    Packets: Sent = 4, Received = 4, Lost = 0 (0% loss),
    Approximate round trip times in milli-seconds:
        Minimum = 13ms, Maximum = 36ms, Average = 22ms

C:\>
```

ping 203.0.113.78 แล้วมีการตอบกลับแสดงว่าถูกต้อง



มีหน้าเว็บไซต์ของ Cisco Packet Tracer โหลดขึ้นมาแสดงผลบนหน้าจอ แสดงว่าถูกต้อง

